

TidyTuesday Week 40

Pranay Gundam

Tuesday 29th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we take two economic time series that have never met, sit them down together, and demand they explain their relationship to us. As always, nobody asked for this, and yet here we are. This week featured just two brave contestants, and both delivered performances I frankly wasn't emotionally prepared for.

First up, on September 28th, we paired Real GDP of Wood Product Manufacturing in the Great Lakes region with, and I cannot stress how naturally these two belong together, Mobile Cellular Subscriptions per 100 People in Europe and Central Asia. The raw regression looked at these two and shrugged: an R-squared of about 0.08 and a p-value of 0.14, which is the statistical equivalent of a mumbled "eh, maybe." But then we detrended, stripping out each series' own march through time, and against all reason the relationship got stronger, not weaker. Suddenly R-squared doubled to 0.18 and the p-value slid under 0.05. So the message is: once you remove the boring shared drift, Great Lakes lumber and European cell phones have a modest, statistically respectable negative wiggle-for-wiggle correlation. Do I have a theory? Yes. Is it that Europeans upgrade their phones in years when Michigan mills feel sad? It is not, because there is no theory, because this is nonsense, but a slightly more robust flavor of nonsense than usual.

Then September 29th arrived and produced the kind of raw regression number that makes you want to lie down. Shares of gross domestic income going to employee wages (a series that has been discontinued, God rest it) versus the number of Black or African American people not in the labor force. The raw R-squared was 0.886. Zero point eight eight six! In an honest project we'd frame this and hang it on the wall. But this is not an honest project, this is the Trend Mirage Detection Bureau, and sure enough, once we detrended both series, that gorgeous 0.89 evaporated down to 0.23. A textbook case of two things drifting through the twentieth century in loose lockstep and pretending it means something. That said, 0.23 with a p-value under 0.002 didn't fully collapse, so there's a faint pulse still detectable after the trend transplant. I'd call it "interesting," in the sense that literally everything here is interesting if you squint hard enough and have no standards.

The usual disclaimer, now with feeling: these are randomly paired series, chosen by a dice-rolling script that has no idea what economics is and even less idea what causation means. Spurious correlation isn't the risk here, it's the entire business model. Wood products do not control European phones, and please, for the love of all that is measurable, do not cite any of this in anything. See you next week for more strangers being forced to hold hands.

2 Date: 2026-09-28

Series ID: GLAKWPMANRGSP

This series is titled Real Gross Domestic Product: Wood Product Manufacturing (321) in the Great Lakes BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: ITCELSETSP2ECS

This series is titled Mobile Cellular Subscriptions: All Income Levels in Europe and Central Asia and has a frequency of Annual. The units are Number per 100 People and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2025-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITCELSETSP2ECS	R-squared:	0.083
Model:	OLS	Adj. R-squared:	0.048
Method:	Least Squares	F-statistic:	2.357
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.137
Time:	18:50:21	Log-Likelihood:	-142.11
No. Observations:	28	AIC:	288.2
Df Residuals:	26	BIC:	290.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	285.3207	123.888	2.303	0.030	30.665	539.976
value_fred_GLAKWPMANRGSP	-0.0407	0.026	-1.535	0.137	-0.095	0.014

Omnibus:	3.492	Durbin-Watson:	0.174
Prob(Omnibus):	0.174	Jarque-Bera (JB):	2.983
Skew:	-0.713	Prob(JB):	0.225
Kurtosis:	2.275	Cond. No.	7.63e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 7.63e+04. This might indicate that there are strong multicollinearity or other numerical problems.

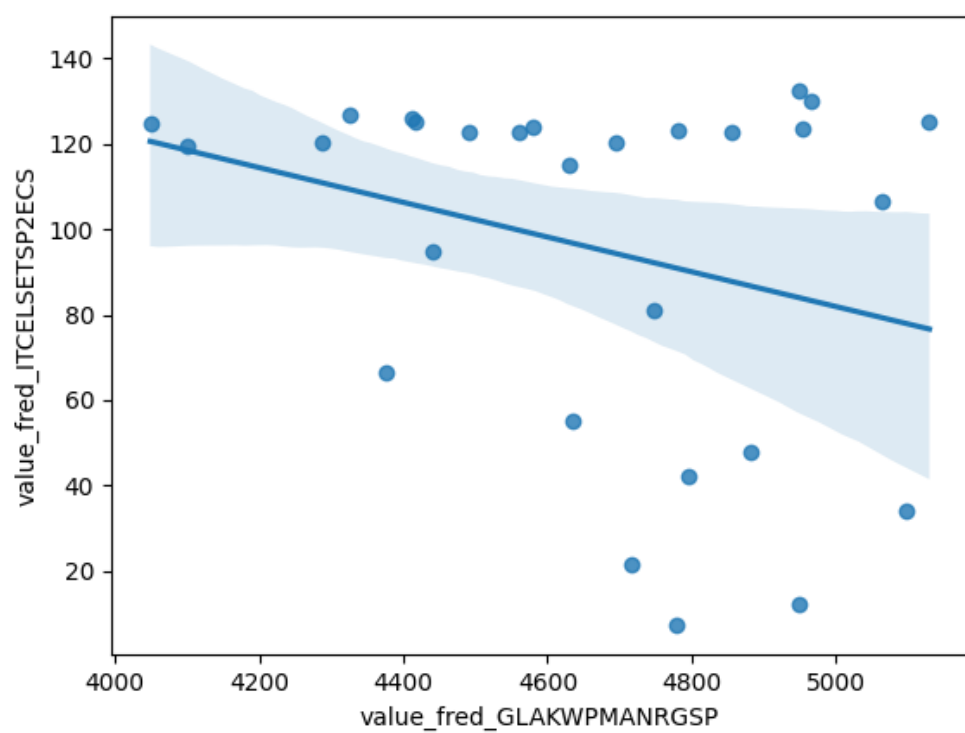


Figure 1: Raw Regression Plot for 2026-09-28

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITCELSETSP2ECS_detrended	R-squared:	0.184
Model:	OLS	Adj. R-squared:	0.152
Method:	Least Squares	F-statistic:	5.853
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.0228
Time:	18:50:21	Log-Likelihood:	-119.03
No. Observations:	28	AIC:	242.1
Df Residuals:	26	BIC:	244.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.116e-12	3.330	3.35e-13	1.000	-6.845	6.845
value_fred_GLAKWPMANRGSP_detrended	-0.0283	0.012	-2.419	0.023	-0.052	-0.004

Omnibus:	0.861	Durbin-Watson:	0.444
Prob(Omnibus):	0.650	Jarque-Bera (JB):	0.837
Skew:	0.216	Prob(JB):	0.658
Kurtosis:	2.272	Cond. No.	285.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

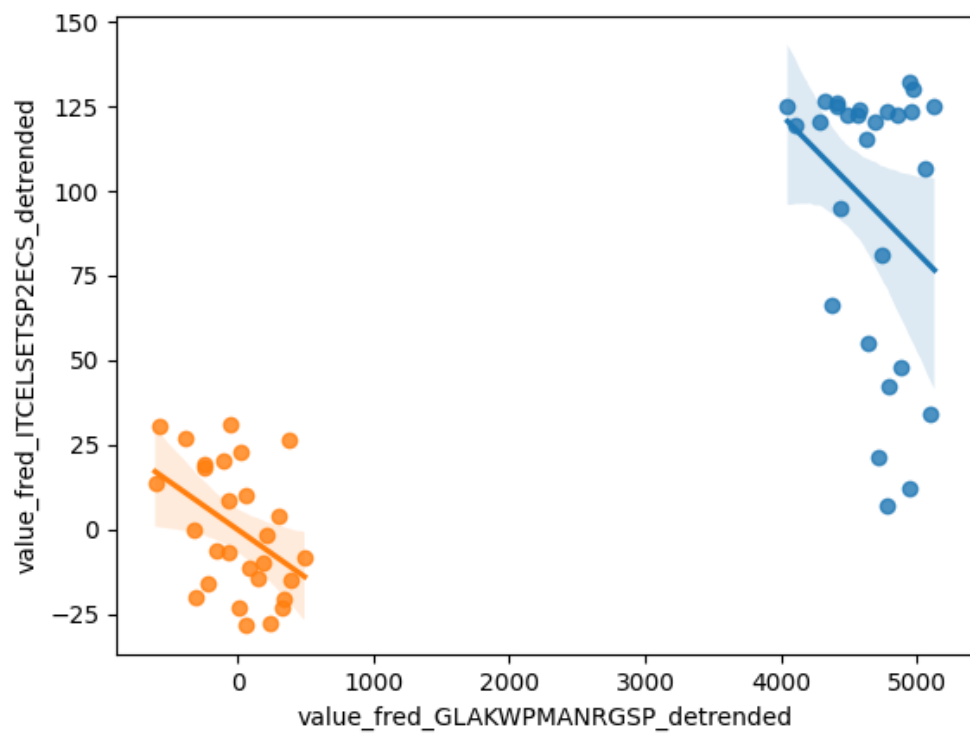


Figure 2: Detrended Regression Plot for 2026-09-28

3 Date: 2026-09-29

Series ID: W269RE1A156NBEA

This series is titled Shares of gross domestic income: Compensation of employees, paid: Wage and salary accruals: Disbursements (DISCONTINUED) and has a frequency of Annual. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2011-01-01. The popularity of this series is 1.

Series ID: LNU05000006

This series is titled Not in Labor Force, Black or African American and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1972-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LNU05000006	R-squared:	0.886
Model:	OLS	Adj. R-squared:	0.883
Method:	Least Squares	F-statistic:	295.5
Date:	Tue, 29 Sep 2026	Prob (F-statistic):	1.63e-19
Time:	17:07:32	Log-Likelihood:	-300.21
No. Observations:	40	AIC:	604.4
Df Residuals:	38	BIC:	607.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	3.833e+04	1754.440	21.850	0.000	3.48e+04	4.19e+04
value_fred_W269RE1A156NBEA	-632.0219	36.768	-17.190	0.000	-706.454	-557.590

Omnibus:	2.321	Durbin-Watson:	0.414
Prob(Omnibus):	0.313	Jarque-Bera (JB):	1.433
Skew:	0.178	Prob(JB):	0.488
Kurtosis:	2.143	Cond. No.	1.17e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.17e+03. This might indicate that there are strong multicollinearity or other numerical problems.

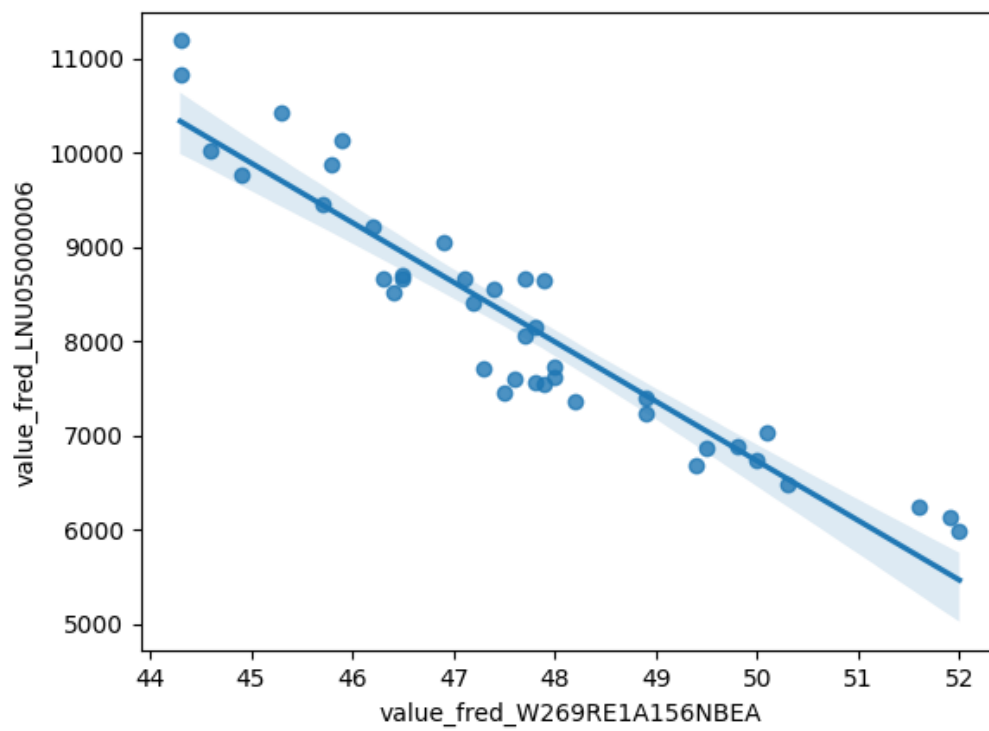


Figure 3: Raw Regression Plot for 2026-09-29

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LNU05000006_detrended	R-squared:	0.229
Model:	OLS	Adj. R-squared:	0.209
Method:	Least Squares	F-statistic:	11.29
Date:	Tue, 29 Sep 2026	Prob (F-statistic):	0.00178
Time:	17:07:32	Log-Likelihood:	-275.80
No. Observations:	40	AIC:	555.6
Df Residuals:	38	BIC:	559.0
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.404e-11	38.755	1.14e-12	1.000	-78.455	78.455
value_fred_W269RE1A156NBEA_detrended	-174.7065	51.991	-3.360	0.002	-279.957	-69.456

Omnibus:	8.056	Durbin-Watson:	0.312
Prob(Omnibus):	0.018	Jarque-Bera (JB):	6.850
Skew:	0.862	Prob(JB):	0.0325
Kurtosis:	4.067	Cond. No.	1.34

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

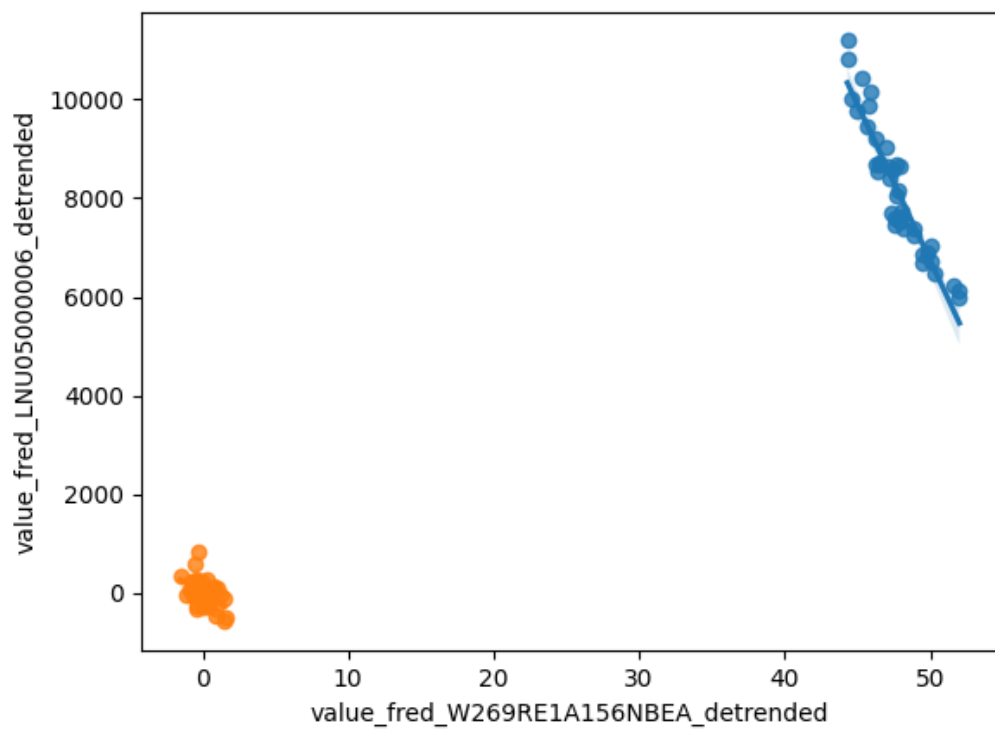


Figure 4: Detrended Regression Plot for 2026-09-29